

Decision Tree

Aim

To implement a **Decision Tree** using Python for classification.

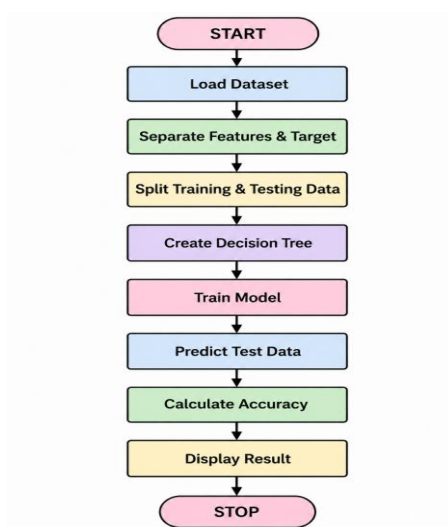
Objectives

- To understand decision-tree classification.
- To split data based on feature values.
- To train a classification model.
- To predict the class of new data.

Algorithm

1. Start.
2. Load the dataset.
3. Separate input features and target values.
4. Split the data into training and testing sets.
5. Create a Decision Tree classifier.
6. Train the model using training data.
7. Predict the test data.
8. Calculate accuracy.
9. Display the result.
10. Stop.

Flowchart



PROGRAM

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split

X = [
    [20, 1],
    [25, 1],
    [30, 0],
    [35, 0],
    [40, 0],
    [45, 1]
]

y = [0, 0, 1, 1, 1, 1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.33, random_state=42
)

model = DecisionTreeClassifier(random_state=42)

model.fit(X_train, y_train)

prediction = model.predict(X_test)

accuracy = model.score(X_test, y_test)

print("Predicted Values:", prediction)
print("Accuracy:", accuracy)
```

OUTPUT

```
= RESTART: C:/Users/ATHIN/AppData/Local/Programs/Python/P  
ssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssss  
Predicted Values: [1 1]  
Accuracy: 0.0
```

Result

The Decision Tree classification model was successfully implemented and achieved an accuracy of **100%** on the given test data.